BugBoard26 - UM L Class Diagram (Backend + Frontend)

backend (it.unina.bugboard)

backend.m odel

backend.utils.enum s

backend.dto

backend.repository

backend.services

b_repo

backend.controller

b_srv

backend.security

frontend (application)

fro n ten d.ap p

fro n ten d.ap i

fro n te n d.d a ta

frontend.view s

f_app

f_d ata f_api

«SpringBootApplication»
BugBoard26BackendApplication

+m ain(args: String[]): void

User

- id: Long
- em ail: String
- psw: String
- role: Role

+getId(): Long
+getEm ail(): String
+getRole(): Role

Issue

- id: Long
- title: S tring
- description: String
- priority: Priority
- type: Type
- state: State

- path: String
- createdAt: DateTime
- updatedAt: DateTime
- creator: User

+getId(): Long
+getTitle(): String
+getState(): State
+setState(state: State): void

«enum»

Role

USER
ADMIN

«enum»

Priority

HIGH
MEDIUM
LOW

«enum»

State

TODO
ONGOING
COMPLETED

«enum»

Type

QUESTION
BUG
DOCUMENTATION
FEATURE

«enum»

AllowedField

PRIORITY(property: String)
STATE(property: String)
CREATED_AT(property: String)

+getProperty(): String

« u tility»
StringManager

+splitPath(path: String): String[]
+splitByDelim iter(s: String, delim: String): String[]

«DTO»

LoginRequest

+em ail: String
+psw: String

«DTO»

LoginResponse

+userID: Long
+role: Role
+token: String

«DTO»

RegisterRequest

+em ail: String
+psw: String
+role: Role

«DTO»

NewIssueRequest

+ title: S tring
+description: String
+type: Type
+priority: Priority?

«DTO»

ModifyRequest

+title: S tring?
+description: String?
+type: Type?
+priority: Priority?
+state: State?

«DTO»

IssueResponse

+id: Long
+ title: S tring
+description: String
+priority: P riority
+state: S tate
+type: Type
+path: S tring
+createdAt: DateTim e
+updatedAt: DateTim e
+creatorId: Long

DatabaseUserInterface

+save(u: User): User
+findById(id: Long): Optional<User>
+findByEm ail(em ail: String): Optional<User>

DatabaseUserRepository

-jpa: JpaUserRepositoryInterface

+save(u: User): User
+findById(id: Long): Optional<User>
+findByEm ail(em ail: String): Optional<User>

<<JpaRepository<User,Long>>>
JpaUserRepositoryInterface

+save(u: User): User
+findById(id: Long): Optional<User>
+findByEm ail(em ail: String): Optional<User>

DatabaseIssueInterface

+save(i: Issue): Issue
+findById(id: Long): Optional<Issue>
+findAll(): List<Issue>

DatabaseIssueRepository

-jpa: JpaIssueRepositoryInterface

+save(i: Issue): Issue
+findById(id: Long): Optional<Issue>
+findAll(): List<Issue>

<<JpaRepository<Issue,Long>>>
JpaIssueRepositoryInterface

+save(i: Issue): Issue
+findById(id: Long): Optional<Issue>
+findAll(): List<Issue>

UserServicesInterface

+login(req: LoginRequest): LoginResponse
+register(req: RegisterRequest): LoginResponse
+findUserById(id: Long): User

UserServices

- users: DatabaseUserInterface
- passwordEncoder: PasswordEncoder

+login(req: LoginRequest): LoginResponse
+register(req: RegisterRequest): LoginResponse
+findUserById(id: Long): User

IssueServicesInterface

+createIssue(req: NewIssueRequest, creatorId: Long): IssueResponse
+getAllIssues(sortBy: AllowedField?): List<IssueResponse>
+modifyIssue(id: Long, req: ModifyRequest, requesterId: Long, isAdmin: boolean): IssueResponse
+uploadIssueImage(issueId: Long, file: File): String
+getIssueImage(issueId: Long): Resource

IssueServices

- issues: DatabaseIssueInterface
- uploadRoot: Path
- publicBase: String

+createIssue(req: NewIssueRequest, creatorId: Long): IssueResponse
+getAllIssues(sortBy: AllowedField?): List<IssueResponse>
+modifyIssue(id: Long, req: ModifyRequest, requesterId: Long, isAdmin: boolean): IssueResponse
+uploadIssueImage(issueId: Long, file: File): String
+getIssueImage(issueId: Long): Resource

DatabaseIssueInterface DatabaseUserInterface PasswordEncoder

«RestController»
UserController

+login(req: LoginRequest): LoginResponse
+adminRegister(req: RegisterRequest): LoginResponse

«RestController»
IssueController

+newIssue(req: NewIssueRequest): IssueResponse
+viewIssues(sortBy: AllowedField?): List<IssueResponse>
+modifyIssue(id: Long, req: ModifyRequest): IssueResponse
+uploadImage(id: Long, file: File): String
+getImage(id: Long): Resource

SecurityContextHolder IssueServicesInterface UserServicesInterface JwtService

SecurityConfig

+securityFilterChain(): SecurityFilterChain
+passwordEncoder(): PasswordEncoder

JwtService

+createToken(userId: Long, role: Role): String
+verify(token: String): boolean
+extractUserId(token: String): Long
+extractRole(token: String): Role

«OncePerRequestFilter»
JwtFilter

+doFilterInternal(req, res, chain): void

OncePerRequestFilter UsernamePasswordAuthenticationToken

«JavaFX Application»
M ain

+start(stage: Stage): void

« u tility»
AppNavigator

- rootStage: Stage

+init(stage: Stage): void
+goLogin(): void
+goDashboard(): void
+goAccount(): void
+goReportIssue(): void
+goAdminCreateUser(): void
+goModifyIssue(item: IssueItem): void
+goIssuesList(): void
- setRoot(view): void

« u tility»
Session

- userId: Long
- role: String
- token: String
- em ail: String

+isAdm in(): boolean
+isLoggedIn(): boolean
+getBearerTokenOrNull(): String
+clear(): void

«config»
ApiConfig

+BASE_URL: String

«client»

AuthApi

+login(baseUrl: String, email: String, psw: String): LoginResponse

«client»

AdminApi

+registerUser(email: String, psw: String, role: String): void

«client»

IssueApi

+getIssues(sort: String): List<IssueItem >
+createIssue(title: String, desc: String, type: String, priority: String?): IssueItem
+modifyIssue(id: Long, title: String?, desc: String?, type: String?, priority: String?, state: String?): IssueItem
+uploadIssueImage(issueId: Long, file: File): void
+downloadIssueImage(url: String): Image
+downloadIssueImageWithFallback(issueId: Long, url: String): Image

«exception»
UnauthorizedException

«exception»
ForbiddenException

«exception»
ConflictException

Java HttpClient

«record»
IssueItem

+id: Long
+ title: S tring
+description: String
+priority: S tring
+state: S tring
+type: S tring
+path: S tring
+createdAt: String
+updatedAt: String
+creatorId: Long

«record»

LoginResponse

+userID: Long
+role: S tring
+token: String

«enum»

TypeOpt

QUESTION
BUG
DOCUMENTATION
FEATURE

«enum»

PriorityOpt

HIGH
MEDIUM
LOW

«View»

LoginView

+onLogin(email: String, psw: String): void

«View»
DashboardView

+render(): void

«View»
AccountView

+render(): void
+logout(): void

«View»
AdminCreateUserView

+createUser(email: String, psw: String, role: String): void

«View»

ReportIssueView

+subm it(title: String, desc: String, type: String, priority: String?): void
+uploadImage(file: File): void

«View»
IssuesListView

+load(sort: String): void
+openDetail(item: IssueItem): void
+openModify(item: IssueItem): void

«View»

ModifyIssueView

+saveChanges(item: IssueItem, title: String?, desc: String?, type: String?, priority: String?, state: String?): void
+uploadImage(file: File): void

«View»
IssueDetailView

+show(item: IssueItem): void
+loadIm age(): void

«View»
PlaceholderView

« u tility»
JwtDebug

+decodeHeader(token: String): String
+decodePayload(token: String): String

AppNavigator Session

LoginResponse IssueItem IssueApi AuthApi AdminApi

creator

@ManyToOne

0..*

1

delegates delegates

uses

uses uses

uses uses

depends depends
depends

depends
reads principal(userId)

uses

sets auth (ROLE_*)

encodes/decodes

in it()

reads role/logged

uses BASE_URL

uses BASE_URL

uses BASE_URL

uses bearer? (none)

uses bearer

uses bearer

uses

goDashboard()

navigate

goLogin()
goModifyIssue(item)

uses

receives

populates

reads/clear isAdmin()

requires adm in

uses

uses

uses auth

creates/receives uses

displays

creates/opens

uses

admin or creator

edits

downloads image

displays

reads current user

reads token

goModifyIssue(item)